



Atmospheric Thermodynamics

Chapter 2: First Law of Thermodynamics



Objective

- To Define the first law of Thermodynamics
- To Know the concept of work
- To Explain Joule's Law
- To Describe the Specific Heats

First Law of Thermodynamics

- In meteorology, thermodynamics plays a crucial role in understanding atmospheric processes, such as energy exchanges, cloud formation, and large-scale weather patterns.
- The **first law of thermodynamics** states that energy is conserved in a system. It can be written as

$$dQ = dU + dW$$

where:

- dQ = heat added to the system (J)
- dU = change in internal energy (J)
- dW = work done by the system (J)

Meteorological Relevance:

- The first law helps explain how air parcels gain or lose heat, affecting their temperature, pressure, and volume.
- It governs atmospheric processes like adiabatic heating and cooling, which is essential in cloud and storm formation.
- For example, when air rises in the atmosphere, it expands and does work on its surroundings, causing it to cool. This is the basis of adiabatic cooling, which leads to cloud formation.

General statement of the law

- ✓ The first law of Thermodynamics is also known as law of conservation of energy.
- ✓ The First Law of Thermodynamics states that heat is a form of energy, and thermodynamic processes are therefore subject to the principle of conservation of energy.
- ✓ This means that heat energy cannot be created or destroyed. It can, however, be transferred from one location to another and converted to one form of energy to other forms of energy.
- ✓ The First Law of Thermodynamics says that the internal energy of a system has to be equal to the work that is being done on the system, plus or minus the heat that flows in or out of the system and any other work that is done on the system.

- ❖ A mass of gas possesses internal energy due to *the kinetic and potential energy* of its molecules or atoms.
 - ✓ Changes in internal energy are manifested as changes in the temperature of the system.
- ❖ Suppose that a closed system of unit mass takes in a certain quantity of thermal energy q , which it can receive by thermal conduction and/or radiation.
 - ✓ As a result the system may do a certain amount of external work w .

- ❖ The excess of the energy supplied to the body over and above the external work done by the body is $q - w$.
- ❖ It follows from the *principle of conservation of energy* that the internal energy of the system must increase by $q - w$. That is,

$$\Delta u = q - w$$

- ✓ where Δu is the change in internal energy of the system.

$$u_2 - u_1 = q - w$$

- ✓ where u_1 and u_2 are the internal energies of the system before and after the change.

- ❖ In differential form this becomes: $du = dq - dw$
 - ✓ where ***dq*** is the differential increment of ***heat*** added to the system,
 - ✓ ***dw*** the differential element of ***work*** done by the system, and
 - ✓ ***du*** the differential increase in ***internal energy*** of the system.
- ❖ This is a statement of the ***First Law of Thermodynamics***.
- ❖ In general, First Law of Thermodynamics embodies three related ideas:
 - i. defines internal energy
 - ii. states that heat is a form of energy.
 - iii. states that energy is conserved

Internal energy (U)

- Internal energy (U) is the total microscopic energy contained within a system.
- It includes: Kinetic energy of molecules (motion, rotation, vibration). Potential energy from intermolecular forces.
- In meteorology, internal energy is crucial for understanding atmospheric temperature changes, stability, and heat exchange in air parcels.

- The first law of thermodynamics expresses the principle of conservation of energy for thermodynamical systems.
- The idea here is that energy cannot be created or destroyed. It can only change from one form to another.

Formulas for Internal energy (U)

- Internal Energy for an Ideal Gas
- For an ideal gas, internal energy depends only on temperature (not on pressure or volume):
- $dU = C_v dT$

$C_v =$ Specific heat at constant volume (J/kg·K)

$dT =$ Change in temperature

Meteorological Relevance:

- The total energy in an air parcel determines its behavior in the atmosphere.
- Hurricanes derive energy from warm ocean waters through latent heat release.
- Atmospheric circulation involves energy transfers between the equator and poles, driving weather systems.

Heat Added to the System (Q)

- Heat (Q) is the transfer of thermal energy into or out of a system due to a temperature difference.
- In meteorology, heat input or loss drives temperature changes, atmospheric convection, cloud formation, and weather patterns.

- $\delta Q = dU + dW$
- This means the heat added to a system is used for:
- Changing internal energy (dU) (increasing temperature).
- Doing work (dW) (expansion of air, for example).

Meteorological Importance of Heat Transfer

❑ Sensible Heat and Temperature Changes

- ✓ Direct heating of the atmosphere from solar radiation or surface conduction increases temperature.

❑ Latent Heat and Phase Changes

- ✓ Heat added during evaporation, condensation, sublimation, and freezing influences cloud and storm formation. Example: Condensation releases heat, powering thunderstorms.

❑ Adiabatic Heating and Cooling

- ✓ In an adiabatic process, temperature changes only due to pressure and volume changes (e.g., rising air cools).

❑ Weather Systems and Circulations

- ✓ Heat exchange between oceans, land, and atmosphere drives wind patterns, monsoons, and hurricanes.

Work

- In meteorology, work is often done by or on air parcels as they expand or contract due to pressure changes. The work done by an expanding gas is:

$$dw = p dv$$

Where

P=pressure(pa)

dv= change in volume(m³)

- Work (W) in thermodynamics refers to energy transfer due to volume change in response to pressure.
- In meteorology, work is critical for understanding air parcel expansion and compression, adiabatic processes, and atmospheric dynamics.

Cont. . .

- If $dV > 0$ (the system expands) it follows that $W > 0$.
- if $dV < 0$ (system contracts) it follows that $W < 0$.
- Thus, positive work corresponds to work done by the system on the environment .
- Negative work corresponds to work done to the system by the environment.

Work Done by the System (Expansion Work)

- When an air parcel expands, it does work on its surroundings by pushing against the surrounding air.
- This occurs when the parcel moves into a lower-pressure region (e.g., rising air in the atmosphere).
- Since energy is used to perform this work, the internal energy of the air parcel decreases, leading to a drop in temperature if no heat is added (adiabatic cooling).

Work Done on the System (Compression Work)

- When an air parcel is compressed, the surroundings do work on the parcel.
- This happens when the parcel moves into a higher-pressure region (e.g., sinking air).
- The energy gained by the parcel increases its internal energy, leading to a temperature rise if no heat is lost (adiabatic warming).

Equivalence Between Heat and Work

- a concept established by James Prescott Joule through his experiments.
- The amount of heat (Q) required to raise 1 gram of water by 1°C (from 14.5°C to 15.5°C) is defined as 1 calorie (cal).

Mechanical Equivalent of Heat

- Joule's experiments showed that heat and mechanical work are equivalent forms of energy.
- He determined that 1 calorie of heat energy is equivalent to 4.185 Joules of mechanical work. This means:

$$1\text{cal} = 4.185\text{ J}$$

Equivalence Between Heat and Work

- In atmospheric thermodynamics, the equivalence between heat and work is a fundamental principle derived from the First Law of Thermodynamics, which states that
- energy can be transferred between a system and its surroundings in the form of heat (Q) and work (W), but the total energy is conserved.

$$\Delta u = q - w$$

Explanation of Equivalence

- ❑ **Interchangeability:** Heat and work are two different ways of transferring energy. If a system gains energy as heat, it can be entirely converted into work under ideal conditions (such as in a Carnot cycle).
- ❑ **Irreversibility and Efficiency:** While heat can be converted into work, the reverse process (work to heat) is often more efficient. For example, frictional processes convert work into heat, but converting all heat into work requires an idealized cycle.

Work $\rightarrow$ Heat (Very Efficient)

- ❑ Converting work into heat is very easy and almost 100% efficient. This usually happens through friction or resistance.

Example:

- Rubbing your hands together $\rightarrow$ mechanical work becomes heat.
- Air turbulence $\rightarrow$ wind energy becomes heat.

Heat $\rightarrow$ Work (Less Efficient)

- Converting heat into work is much more difficult.
- It requires a heat engine, and according to the Second Law of Thermodynamics, not all heat can be converted into work.
- Some heat must always be lost to a colder environment.

Example:

- A steam engine
 - A power plant
 - A car engine
- These systems convert heat energy into mechanical work, but their efficiency is always less than 100%.

Form of the first law in the following special cases

- **Isothermal transformations:** $i \xrightarrow{T=\text{constant}} f$

During such transformations $dT = 0$ and as such from equation () it follows that $\delta Q = \delta W$. Then the amount of heat exchanged is

$$Q = \int_i^f \delta W = \int_i^f p dV$$

or

$$Q = nR^*T \int_i^f \frac{dV}{V}$$

or

$$Q = nR^*T \ln \frac{V_f}{V_i}.$$

- **Isochoric transformations:** $i \xrightarrow{V=\text{constant}} f$

In this case $dV = 0$ and from the first law it follows that

$$\delta Q = dU = C_V dT$$

or

$$\Delta U = Q = C_V \int_i^f dT = C_V (T_f - T_i).$$

- **Isobaric transformations:** $i \xrightarrow{p=\text{constant}} f$

in this case it follows that

$$\delta Q = C_p dT \quad \text{or} \quad Q = C_p (T_f - T_i),$$

$$\delta W = p dV \quad \text{or} \quad W = p(V_f - V_i).$$

Example

One mole of a gas expands from a volume of 10 liters and a temperature of 300 K to:

(a) A volume of 14 liters and a temperature of 300 K, and

What is the work done by the gas on the environment in each case?

Special Cases of Work in Meteorology

Constant Volume Process

Constant Pressure Process ($P=\text{constant}$)

C. Adiabatic Process ($Q=0$)

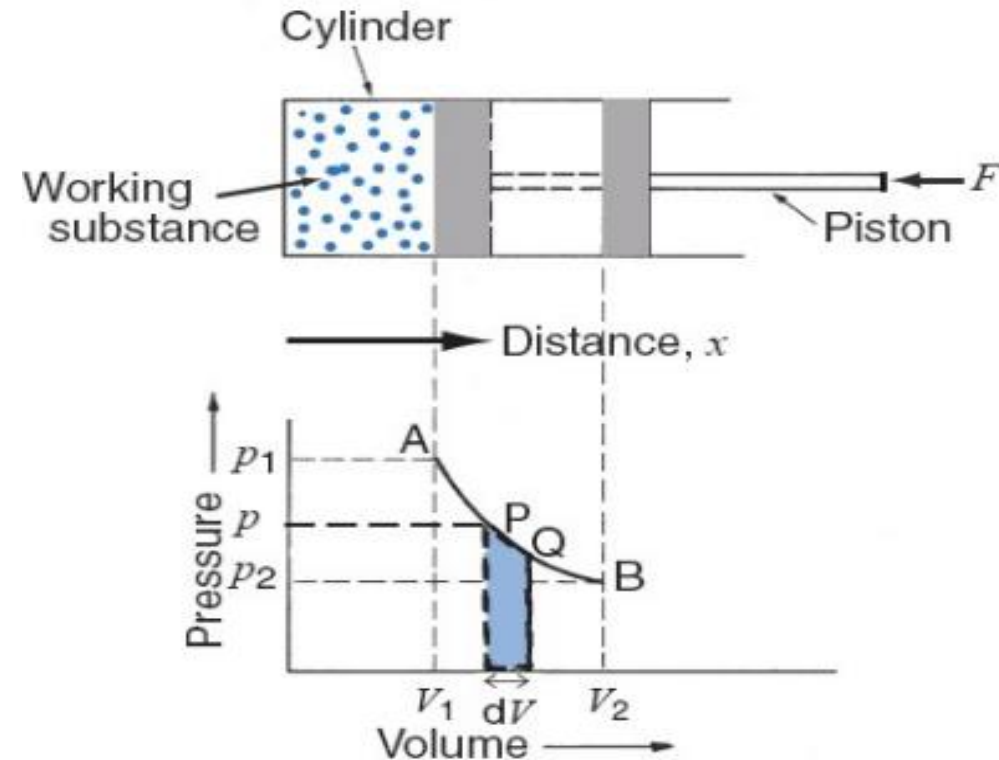
The concept of work

- $W = Fdx$, $P = F/A \leftrightarrow F = PA$
- $dW = PAdx = pdV$ (shaded region of the graph)
- Specific work (work per unit mass): $dw = p d\alpha$
- Work is found by integrating this differential over the initial and final volumes V_1 and V_2 :

$$W = \int_{V_1}^{V_2} p dV$$

W = area under the curve AB

- ❖ If $V_2 > V_1$, W is positive, the substance does work on its environment.
- ❖ If $V_2 < V_1$, then W is negative, the environment does work on the substance.



This thermodynamic diagram (p-V) represents the state of the system at every point along the line. (Recall that the p-V diagram is a simple thermodynamic diagram that define the physical states of the matter by only two thermodynamic variables (P & V.)

❖ The thermodynamic equation may be written

$$dq = du + dw$$

❖ Using $dw = pd\alpha \rightarrow dq = du + pd\alpha$

❖ Which is an alternative statement of the First Law of Thermodynamics.

Example

- Calculate the work done in compressing (isothermally) 2 kg of dry air to one-tenth its initial volume at 15 °C.

Hint: Use the ideal gas law and work concept in combination.

Joule's Law

- Joule's Law states that the internal energy of an ideal gas depends only on temperature, not pressure or volume:

Meteorological Relevance

- Joule's law explains why changes in temperature affect the energy content of air masses.
- It helps predict temperature changes in adiabatic expansion/compression, key in understanding atmospheric stability.

Thermal capacities (i.e Specific Heats)

- ❖ **Heat capacity (C)** is the amount of energy needed to raise the temperature of a specific substance by 1 degree Celsius.
- ❖ Heat capacity can also be viewed as the ratio of the amount of energy transferred to an object and the resultant temperature rise (ΔT).

$$C = \frac{Q}{\Delta T}$$

- ❖ Suppose a small quantity of heat dq is given to a unit mass of a material and as a consequence, the temperature of the material increases from T to $T + dT$ without any changes in phase occurring within the material.
- ❖ The ratio dq/dT is called the ***specific heat*** of the material.

Air behaves differently in dry vs. moist conditions:

- Moist air has a higher heat capacity due to the presence of water vapor, affecting storm dynamics.
- Oceanic heat capacity influences climate by storing and releasing heat over long periods.

- ❖ If the *volume* of the material is kept constant, a *specific heat at constant volume*, c_v , is defined

$$c_v = \left(\frac{dq}{dT} \right)_v$$

- ❖ But if the volume of the material is constant, the thermodynamic equation gives $dq = du$. Therefore

$$c_v = \left(\frac{du}{dT} \right)_v$$

- ❖ For an ideal gas, Joule's law applies and therefore u depends only on temperature. Therefore, regardless of whether the volume of a gas changes, we may write

$$c_v = \frac{du}{dT}$$

❖ In differential form, we have:

$$du = c_v dT$$

❖ Then, the First Law of Thermodynamics for an ideal gas can be written in the form

$$dq = c_v dT + p d\alpha$$

❖ Since u is a function of state, no matter how the material changes from state 1 to state 2, the change in its internal energy is

$$u_2 - u_1 = \int_{T_1}^{T_2} c_v dT$$

❖ We can also define a *specific heat at constant pressure*

$$c_p = \left(\frac{dq}{dT} \right)_p$$

❖ Where the material is allowed to expand as heat is added to it and its temperature rises, but its pressure remains constant.

❖ In this case, some of the heat added to the material will have to be expended to do work as the system expands against the constant pressure of its environment.

❖ Therefore, a larger quantity of heat must be added to the material to raise its temperature by a given amount than if the volume of the material were kept constant.

❖ Therefore

$$c_p > c_v$$

❖ For the case of an ideal gas, this inequality can be seen mathematically as follows. We write the thermodynamic equation as

$$dq = c_v dT + p d\alpha = c_v dT + d(p\alpha) - \alpha dp$$

❖ From the equation of state, $d(p\alpha) = d(RT) = RdT$.

❖ Therefore,

$$dq = (c_v + R)dT - \alpha dp$$

❖ At constant pressure, the last term vanishes; therefore,

$$c_p = \left(\frac{dq}{dT} \right)_p = c_v + R$$

❖ Using this in the equation above it, we obtain an alternative form of the First Law of Thermodynamics:

$$dq = c_p dT - \alpha dp$$

Derivation of thermodynamic energy equation

$$\frac{dq}{dT} = c_p \frac{dT}{dt} - V \frac{dP}{dt}$$

This is thermodynamic energy equation other form of first laws of thermodynamics

✓ Refer to the detailed step-by-step derivation on the classroom board.

Why is This equation is Important?

- This form of the equation is useful because:
 - ✓ It expresses heat transfer (dq) in terms of enthalpy changes.
 - ✓ It is commonly used in atmospheric thermodynamics, where pressure changes (dp) occur frequently.
 - ✓ It shows the connection between heat capacity at constant pressure (cp) and specific volume (α).

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Thanks !

Atmospheric Thermodynamics

Chapter 2: First Law of Thermodynamics



Objective

- To Define the Enthalpy
- To explain lapse rates and Identify It's type
- To Derive Poisson's equation

Enthalpy (H)

- Enthalpy (H) is a thermodynamic quantity that represents the total heat content of a system.
- It is a useful state function when analyzing processes at constant pressure and is especially important in meteorology and atmospheric science because many processes (e.g., phase changes, cloud formation) occur at constant pressure.
- One of the key uses of enthalpy in meteorology is to describe phase changes (e.g., liquid to gas, solid to liquid).

Enthalpy

- **Enthalpy (H)** is the total heat content of a system:

$$H = U + PV$$

where:

- H = enthalpy (J)
- U = internal energy (J)
- P = pressure (Pa)
- V = volume (m³)

- Enthalpy is used primarily in situations where heat exchange occurs at constant pressure, such as during phase transitions or heat flow into or out of a system.
- At constant pressure, the heat added to the system is directly related to the change in enthalpy:

$$dQ=dH$$

- This means that if a system undergoes a process at constant pressure, the amount of heat added or removed from the system equals the change in enthalpy.

Meteorological Relevance

- Enthalpy determines heat transfer in the atmosphere, such as latent heat release during condensation.
- In cloud formation, the enthalpy change drives weather phenomena like thunderstorms and cyclones.

Example:

- When water vapor condenses in clouds, latent heat is released, increasing enthalpy, which fuels storm development.

- ❖ Hence, it becomes convenient to introduce a new function of state called the enthalpy, H , defined by

$$H \equiv U + pV$$

- ❖ Suppose heat is added to a unit mass of material at constant pressure. Suppose the resulting expansion causes the (specific) volume to increase from α_1 to α_2 .

- ❖ Then the work done by a unit mass of the material is
$$\int_{\alpha_1}^{\alpha_2} p d\alpha = p(\alpha_2 - \alpha_1)$$

❖ Therefore, the finite quantity of heat Δq added is given by

$$\Delta q = (u_2 - u_1) + p(\alpha_2 - \alpha_1) = (u_2 + p\alpha_2) - (u_1 + p\alpha_1)$$

❖ where u_1 and u_2 are, respectively, the initial and final internal energies for a unit mass of the material.

Therefore, at constant pressure,

$$\Delta q = h_2 - h_1 = \Delta h$$

❖ where h is the *enthalpy* of a unit mass of the material, which is defined by

$$h \equiv u + p\alpha$$

- ❖ Because u , p , and α are functions of state, h is a function of state. Differentiating the above equation, we obtain

$$dh = du + d(p\alpha)$$

- ❖ We know that $du = c_v dT$ and $dq = c_v dT + d(p\alpha) - \alpha dp$,

- ❖ Replacing $c_v dT = du$, then

$$dq = du + d(p\alpha) - \alpha dp$$

- ❖ Replacing: $du + d(p\alpha) = dh$

$$dq = dh - \alpha dp$$

- ❖ *This is another form of the First Law of Thermodynamics in terms of enthalpy.*

Lapse Rate (Γ)

❖ Lapse rate is the rate at which Earth's atmospheric temperature decreases with an increase in altitude.

➤ Types of lapse rates

❖ There are three types of lapse rates that are used to express the rate of temperature change with a change in altitude,

- ✓ Dry adiabatic lapse rate,
- ✓ Wet adiabatic lapse rate and
- ✓ Environmental lapse rate

Environmental Lapse Rate (ELR) – Real Atmosphere

- It is the actual temperature decrease with altitude measured in the atmosphere at a specific place and time.

Why is it "real"?

- It is measured using weather balloons (radiosondes) or remote sensing.
- It varies depending on location, time, weather, and surface conditions.
- It reflects the natural temperature profile of the atmosphere.
- $6.5^{\circ}\text{C}/\text{km}$

Dry Adiabatic Lapse Rate (DALR) – Theoretical Process

- The rate at which a rising dry air parcel cools without exchanging heat with its surroundings (adiabatic process).
- Why is it "not real" in the same sense as ELR?
 - It applies only to an isolated (idealized) parcel of air that is rising or sinking.
 - It assumes no heat exchange with the environment.
 - The entire atmosphere does not necessarily follow this rate.
 - $9.8^{\circ}\text{C}/\text{km}$

Moist Adiabatic Lapse Rate (MALR) – Also a Theoretical Process

- The rate at which a rising saturated air parcel cools, taking into account the latent heat release from condensation
- Why is it also "not real" like ELR?
 - It applies to an idealized saturated air parcel, not the whole atmosphere.
 - The value varies because the amount of moisture in the air affects how much latent heat is released.
 - Like DALR, the atmosphere does not necessarily follow this exact rate everywhere
 - 4-7°C/km

Why Does This Matter?

- Meteorologists compare ELR with DALR & MALR to determine atmospheric stability.
- If $ELR > DALR$, the atmosphere is unstable (air rises easily $\rightarrow$ thunderstorms).
- If $ELR < MALR$, the atmosphere is stable (air resists rising $\rightarrow$ clear skies)

Difference Between Isothermal Process and Adiabatic Process

Isothermal Process	Adiabatic Process
Transfer of heat occurs in this process.	No heat goes inside or leaves the system.
At a given volume of the substance, the pressure remains high.	At a given volume of the substance, the pressure remains low.
In an isothermal process, the temperature remains invariant.	The temperature varies because of the internal system changes.
Heat can be added or released to the system to keep the temperature constant.	There is no addition of heat nor the release of the heat because maintaining constant temperature doesn't matter here.
The isothermal process has a slower transformation flow.	The adiabatic process has a faster transformation flow.
because of the change in the net heat content of the system.	because of the change in internal energy.

The Dry Adiabatic Lapse Rate

- ❖ We will now derive an expression for the rate of change of temperature with height of a parcel of dry air as it moves about in the Earth's atmosphere.
- ❖ Since the air parcel undergoes only adiabatic transformations ($dq = 0$), and the atmosphere is in hydrostatic equilibrium, for a unit mass of air in the parcel we have:

Cont. . .

$c_v dT + p d\alpha = dq$, but $dq = 0$, **from 1st law of TD & adiabatic**

$c_v dT + d(p\alpha) - \alpha dp = 0$, **from 1st law, @cst V**

$c_v dT + d(RT) - \alpha dp = 0$, **from $d(p\alpha) = d(RT)$**

$(c_v + R)dT + g dz = 0$, **from $\frac{\partial p}{\partial z} = -g\rho$: and $\rho = 1/\alpha$**

$c_p dT + g dz = 0$, **from $c_p = \left(\frac{dq}{dT}\right)_p = c_v + R$**

❖ Dividing both side through dz and c_p simultaneously, we obtain

$$-\left(\frac{dT}{dz}\right) = \frac{g}{c_p} \equiv \Gamma_d$$

❖ where Γ_d is called the dry adiabatic lapse rate.

Cont. . .

❖ Since an air parcel expands as it rises in the atmosphere, its temperature will decrease with height so that Γ_d is a positive quantity.

❖ Substituting $g = 9.81 \text{ m s}^{-2}$ and $c_p = 1004 \text{ J K}^{-1} \text{ kg}^{-1}$ gives

$$\Gamma_d = \frac{g}{c_p} = 0.0098 \text{ K m}^{-1} = 9.8 \text{ K km}^{-1} \approx \textbf{-10 K km}^{-1}$$

✓ Which is the dry adiabatic lapse rate and negative sign indicates the decrement of temperature with increasing altitude (for the case of Tropopause).

Poisson equation and Potential temperature (θ)

- ❖ Potential temperature is defined as the temperature that a parcel of air would have if it were brought adiabatically to some reference pressure.
- ❖ For adiabatic processes, potential temperature is a conserved property of air, which makes it worth keeping track of for many theoretical and some practical purposes.
- ❖ We will derive an expression for the potential temperature of an air parcel in terms of its pressure p , temperature T , and the standard pressure p_0 .

Cont. . .

- ❖ For an adiabatic transformation ($dq = 0$) the thermodynamic equation is

$$c_p dT - \alpha dp = 0$$

- ❖ Using the gas equation $p\alpha = RT$ yields

$$c_p dT - \frac{RT}{p} dp = 0$$

$$c_p dT = \frac{RT}{p} dp$$

$$\frac{c_p}{R} \frac{dT}{T} = \frac{dp}{p}$$

Integrating upward from p_0 (where, by definition, $T=\theta$) to p , we obtain

$$\frac{c_p}{R} \int_{\theta}^T \frac{dT}{T} = \int_{p_0}^p \frac{dp}{p}$$

$$\frac{c_p}{R} \ln \frac{T}{\theta} = \ln \frac{p}{p_0}$$

Cont. . .

❖ Taking the exponential (antilog) of both sides

$$\frac{T}{\theta} = \left(\frac{p}{p_o} \right)^{c_p/R}$$

❖ Solving for θ ,

$$\theta = T \left(\frac{p_o}{p} \right)^{R/c_p}$$

❖ Defining the thermodynamic constant $\kappa = R/c_p$, we get

$$\theta = T \left(\frac{p_o}{p} \right)^{\kappa}$$

❖ ***This equation is called Poisson's equation***

Cont. . .

❖ For dry air, $R = R_d = 287 \text{ J K}^{-1}\text{kg}^{-1}$ and $c_p = 1004 \text{ J K}^{-1}\text{kg}^{-1}$.

❖ Recall that, for a diatomic gas, $R : c_p = 2 : 7$, so

$$\kappa = \frac{2}{7} \approx 0.286$$

❖ Therefore, for dry air the value of $\kappa = 0.286$

Home work

- ❖ A dry air parcel has a temperature of -51°C at the 250 hPa level ($p_o=1000$ hPa)
- a) What is its potential temperature in Kelvin?
 - b) What temperature will the parcel have if it is brought into the cabin of a jet aircraft and compressed adiabatically to a cabin pressure of 850 hPa?

Cont. . .

❖ We can write the thermodynamic equation as:

$$\frac{dq}{dt} = \frac{c_p T}{\theta} \frac{d\theta}{dt}$$

❖ The potential temperature is constant for adiabatic flow.

❖ The entropy is constant for adiabatic flow.

❖ Parameters that remain constant during certain transformations are said to be ***conserved***.

Poisson's equation is important because it:

- Defines potential temperature
- Helps analyze stability
- Describes adiabatic processes
- Is widely used in weather forecasting and models

Summary Table

Concept	Equation	Meteorological Importance
First Law of Thermodynamics	$dQ = dU + dW$	Explains energy conservation in atmospheric processes.
Work	$dW = PdV$	Air parcels expand and cool or compress and warm.
Energy	Various forms	Determines motion of air masses and weather patterns.
Equivalence of Heat & Work	$1 \text{ cal} = 4.186 \text{ J}$	Converts between heat and mechanical energy in the atmosphere.
Heat Capacity	$c = dQ/dT$	Affects atmospheric stability and ocean-atmosphere interactions.
Joule's Law	$U = cvT$	Internal energy changes impact air temperature and stability.
Enthalpy	$H = U + PV$	Latent heat release drives storms and convection.

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Thanks !